

1. Levels of Organization of Life

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Hierarchical Structure

1. Cell: Smallest basic unit of life (e.g. Amoeba, Paramecium).
2. Tissue: Group of similar cells performing a specific function (e.g. xylem, blood).
3. Organ: Group of tissues working together (e.g. leaf, heart).
4. System: Group of organs functioning together (e.g. digestive, circulatory system).
5. Organism: The complete living entity.

Cell Theory

Formulated by Schleiden and Schwann:

iGrades 1½ All living things are composed of one or more cells.

iGrades 1½ The cell is the structural and functional unit of life.

iGrades 1½ All cells arise from pre-existing cells.

2. Plant vs Animal Cells & Organelles

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Key Organelles & Their Functions

iGrades 1/2 Nucleus: Controls cellular activities and houses genetic material (DNA).

iGrades 1/2 Mitochondria: Powerhouse of the cell; site of ATP generation via respiration.

iGrades 1/2 Ribosomes: Sites of protein synthesis.

iGrades 1/2 Chloroplasts: Contain chlorophyll for photosynthesis (plants only).

Differences Between Plant and Animal Cells

iGrades 1/2 Plant Cells: Possess rigid cellulose cell wall, large central vacuole, and chloroplasts.

iGrades 1/2 Animal Cells: Lack cell walls and chloroplasts; possess centrioles and small scattered vacuoles.